Fiscal Year:	FY 2020	**Task Last Updated:**	FY 10/15/2025
PI Name:	Cunha, Micaela Ph.D.		
Project Title:	A Mechanistic Framework to Assess the Efficacy of Aspirin and Other Radio Protectors to Reduce Carcinogenesis by Space Radiations		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
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Space Biology Special Category:	None		
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Zip Code:	10032-3725	**Congressional District:**	13
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Project Type:	Ground	**Solicitation / Funding Source:**	2017 TRI-RFA-17-01: Translational Research Institute for Space Health (TRISH) Postdoctoral Fellowships
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 5/31/2019 per E. Urquieta/TRISH; original end date was 12/31/2019 (Ed., 5/29/19)		
Key Personnel Changes/Previous PI:			
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Task Description:	POSTDOCTORAL FELLOWSHIP NASA is planning a 2-3-year inter-planetary mission to start around 2030, as well as subsequent Mars landing missions. Current data from human and animal studied suggests that exposure of astronauts to radiation in space, in particular to high linear energy transfer (LET) galactic cosmic rays (GCR) and neutrons, may result in increased cancer risks which are not yet adequately quantified. Thus, it is important to develop effective and safe biomedical countermeasures to minimize these risks. One possibility is to use drugs that have been shown to reduce the background spontaneous cancer risks, such as aspirin for gastrointestinal (GI) cancers. We have previously developed a mechanistic framework to evaluate the risk of radiation carcinogenesis and have successfully applied it, taking into consideration multiple variables such as low- and high-LET radiation, low and high dose rates, or age-at-exposure effects. The aim of this project is to extend this framework to assess the effects of biomedical countermeasures on GCR-induced cancer risks. We will start by analyzing data regarding aspirin and GI cancers, as there is convincing evidence that aspirin reduces the risk of colorectal cancer, but the ultimate goal of the proposed project is to provide a general methodology for the assessment of any anti-cancer agent under consideration for reducing the risks of GCR-induced carcinogenesis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Astronauts in space are chronically exposed to low doses of a variety of densely ionizing radiations. On a long interplanetary mission such as on a journey to Mars, the risks to human health of exposure to the space radiation environment are currently the limiting factor, in particular for women and younger astronauts. However, the cellular mechanisms that are involved in the measurable biological response to this peculiar radiation environment are not well known yet, making it difficult to estimate the risks to human health, especially in terms of cancer and other late effects. Developing a mechanistic framework of radiation-induced carcinogenesis that integrates microdosimetric quantities to assess the potential radioprotective effect of chemical compounds will give an important contribution to expand the knowledge on the mechanisms that underlie the biological response to space radiation and will pave the way to more accurate predictions of cancer risks for astronauts.
Task Progress:	Long duration spaceflights, such as to Mars, will involve multiple gravitational transitions that will result in many sensorimotor related difficulties that could jeopardize the mission. Our previous work shows that when subjects are deprived of gravitationally dependent vestibular and somatosensory cues, such as in low-g, 0g, and spaceflight analog environments, they easily become spatially disoriented and show poor learning and performance in an attitude stabilization task. In these experiments we secured subjects into a Multi-Axis Rotation System (MARS) device that was programmed to behave like an inverted pendulum, and participants were instructed to use an attached joystick to stabilize around the balance point. We created spaceflight analog conditions by having subjects dynamically balance in the Horizontal Roll Plane and in the Vertical Yaw Axis, where they did not tilt relative to the gravitational vertical and therefore could not use gravitational cues to determine their position and had to rely only on motion cues. In the first specific aim, our objective was to create an effective training program that would enhance performance and learning in our spaceflight condition. Our training program was based on our previous finding that balance control consists of two dissociable processes: aligning with the gravitational vertical using graviceptive position cues and dynamic stabilization using motion cues. Because subjects in spaceflight conditions can only rely on motion cues, our training program focused on refining subjects' abilities to dynamically stabilize independent of aligning with the gravitational vertical. We did this by setting the balance point to be different from the gravitational vertical in the Vertical Roll Plane. Subjects were instructed to search and then dynamically balance around this balance point. They were not told the location of the balance point and had to ignore biasing positional information from the gravitational cues and instead rely on motion cues to determine the location of the balance point. We found that subjects in this training group, who were later tested in our spaceflight analog condition (Horizontal Roll Plane), showed significant transfer of learning across many metrics, whereas the control group, who balanced in the Horizontal Roll Plane without any training, showed very minimal learning transfer. We also found that the training group retained this learning when retested 4 months later. Subjects that received the training through a visual simulation did not acquire any enhancement in performance when tested in the Horizontal Roll Plane. These findings address NASA roadmap gaps: SM24 and Train-04. In the second specific aim, our objective is to use multiple tests of increasing complexity to characterize and predict the sensorimotor bases of individual differences in our spaceflight analog task. In collaboration with Dr. Faisal Karmali, we will measure semicircular canal thresholds to determine each subject's vestibular sensitivity. Preliminary data, by Karmali and colleagues, reveals a correlation between vestibular thresholds and proficiency in a manual control nulling task to counteract imposed accelerations. However, it is unknown whether vestibular thresholds alone will predict performance in our spaceflight analog task. In the second test, we use a novel experimental paradigm to determine a spatial perception threshold for every subject by blindfolding them and asking them to indicate the balance point as they experience passive motion playback of another subject's dynamic roll balancing performance. To find the threshold, we will alter the magnitude of relevant gravitational cues they receive (by altering the roll plane pitch orientation) and determine at which point they can no longer accurately indicate the balance point. This test allows us characterize individual differences by examining both the acuity of the sensory organs but also the processing of the vestibular signals, such as integration of signals from the semicircular canals to make an estimate of spatial position.
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